

ARUN ARUNACHALAM

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PROFESSIONAL SUMMARY

Senior Data and AI Application Engineer with 17+ years architecting high-performance, cloud-native data platforms and AI-enabled systems across capital markets and enterprise environments at Bank of America. Deep expertise in building GenAI platforms (RAG pipelines, agentic AI), event-driven data engineering (Kafka, Spark, Databricks), and distributed systems that process \$1T+ daily at 99.97% uptime. Proven track record delivering AI-assisted engineering workflows, ML-based intelligence systems, and large-scale data infrastructure modernization that drives measurable business outcomes across demanding, high-throughput environments.

SIGNATURE IMPACT

- **\$8M** enterprise transformation → **40% performance improvement, 30% cost reduction**
- **AI-assisted engineering** → **35% faster delivery, 45% productivity gain**
- Scaled platforms from 10 to **50+ systems** with **99.97% uptime**
- Enabled **~\$2B capital efficiency** via risk platform transformation

CORE CAPABILITIES

AI & GenAI Platforms (RAG, Agentic AI) | Data Engineering (Spark, Databricks, ETL/ELT) | Event-Driven Systems (Kafka) | Cloud-Native Architecture (AWS, Kubernetes) | Distributed & High-Performance Systems | ML Anomaly Detection & Operational Intelligence | AI-Assisted SDLC | Trading & Risk Platforms | Platform Modernization | Regulatory & Compliance Systems | Executive Stakeholder Leadership

PROFESSIONAL EXPERIENCE

Bank of America — Director, Global Risk & Capital Markets Technology (2015 - Present)

- Architected and delivered cloud-native, event-driven data platforms (AWS, Kafka, Kubernetes) supporting **\$1T+ daily processing** across trading and risk systems
- Built GenAI and AI-assisted development workflows, including ML-based anomaly detection, improving delivery speed by **35%** and productivity by **45%**
- Designed and modernized large-scale data engineering pipelines (Spark, Databricks, ETL/ELT, Snowflake) enabling real-time analytics and operational intelligence
- Delivered high-performance, distributed trading, pricing, and risk platforms improving throughput by **40%** at **99.97% uptime**
- Modernized legacy infrastructure into cloud-native, event-driven architecture, scaling from 10 to **50+ integrated systems**
- Ensured regulatory compliance (FRB, FINRA, SEC) across mission-critical, high-availability data systems
- Led global engineering teams (30+ across US/UK/India) driving **15% velocity improvement** through Agile and AI-assisted SDLC practices

Bank of America — VP, Structured Products & Mortgage Technology (2011 - 2015)

- Built real-time pricing and analytics data platforms for structured products and mortgage trading, contributing to **~\$2B capital efficiency**
- Led consolidation of risk data systems, improving data quality, consistency, and downstream analytical performance
- Delivered high-throughput data pipelines supporting complex financial instrument analytics and reporting

Bank of America — AVP, Whole Loan Trading Desk Support (prior to 2011)

- Architected and delivered end-to-end Java EE trading support applications with robust data integration layers
- Introduced CI/CD practices reducing release cycle from monthly to weekly, accelerating data platform delivery cadence

Senior Programmer Analyst — Knight Trading (2004)

DB2 / WebSphere Administrator & Sr. Java Developer — Kyocera Mita (2003)

Sr. Java Web Developer — Mellon Investor Services (2001)

Sr. Java GUI Developer — Thomson Financials (2001)

Sr. Java GUI Developer — Icor Brokerage (2000)

AI & INNOVATION

- Building GenAI platforms including RAG pipelines and agentic AI systems for real-world, high-throughput applications
- Applied ML for anomaly detection and operational intelligence across large-scale distributed data environments
- Driving AI-assisted SDLC workflows that measurably improve delivery speed, code quality, and engineering productivity

EDUCATION

Bachelor of Technology — Anna University

SKILLS

AI & Data: GenAI (RAG pipelines, agentic AI), ML anomaly detection, AI-assisted SDLC, Spark, Databricks, ETL/ELT, Snowflake, Kafka, Hadoop, Cassandra, MongoDB **Architecture:** Cloud-native architecture (AWS, Kubernetes), distributed systems, event-driven systems (Kafka), high-availability platforms (99.97% uptime), enterprise architecture **Engineering:** Python, Java, Node.js, SQL, Oracle, Sybase, REST/SOA, WebLogic, Kubernetes, AWS **Leadership:** Executive stakeholder management, global team leadership, P&L ownership, strategic planning, program management, SDLC, Agile, ITIL, SOX, FRB/FINRA/SEC compliance **Tools:** ServiceNow, JIRA, Git, Visio, Microsoft Project